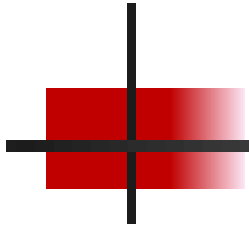
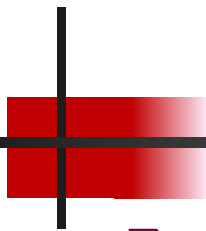


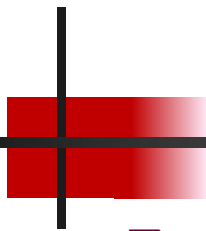
Cyber Security Terminology : Virus, Worm, Trojan Horse, Malware, Ransomware





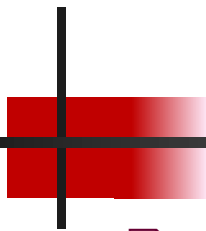
Threats Definition

- ❑ A **threat** is any potential danger to information or systems. It can result in unauthorized access, data theft, corruption, or service disruption.
- ❑ **Types of Threats:**
 - **Passive Threats** – Eavesdropping or monitoring (e.g., traffic analysis, wiretapping).
 - **Active Threats** – Attempt to alter system resources or affect operations (e.g., modification, denial of service).
- ❑ **Categories:**
 - **Natural** – Earthquakes, floods affecting infrastructure.
 - **Human** – Hacking, phishing, social engineering.
 - **Environmental** – Power failure, overheating of systems.



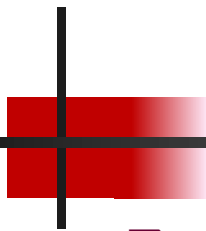
Intruders

- ❑ An **intruder** is an individual or program that attempts to gain unauthorized access to systems or data.
- ❑ **Types of Intruders:**
 - **Masquerader** – **Pretends** to be an authorized user.
 - **Misfeasor (Insider Attack)** – **Authorized user** misuses access.
 - **Clandestine User** – Gains **supervisory control** without authorization.
- ❑ **Attack Techniques:**
 - Password cracking
 - Privilege escalation
 - Spoofing or impersonation



Viruses

- ❑ A **computer virus** is a piece of malicious code that attaches to legitimate files and replicates when the infected file is executed.
- ❑ **Characteristics:**
 - Requires host file/program
 - Spreads through user interaction
 - Can damage files, corrupt systems, or steal information
- ❑ **Types of Viruses:**
 - **File Infector Virus** – Infects executable files.
 - **Boot Sector Virus** – Infects system startup sectors.
 - **Macro Virus** – Targets applications like MS Word or Excel.
 - **Polymorphic Virus** – Changes its code to escape detection.



Worms

- ❑ A **worm** is a standalone malicious program that replicates itself to spread across networks, usually without user interaction.
- ❑ **Key Differences from Virus:**
 - Does **not require** a host file
 - Exploits network vulnerabilities
 - Can cause **network congestion** and system slowdowns
- ❑ **Famous Example:**
 - **ILOVEYOU Worm, WannaCry Ransomware Worm**

Malicious Software (Malware)

- Malware is a general term for software designed to damage, disrupt, or gain unauthorized access to systems.

- Types of Malware:**

Type	Description
Virus	Requires host to replicate
Worm	Standalone, self-replicating
Trojan Horse	Disguised as legitimate software
Spyware	Secretly monitors user activity
Adware	Displays unwanted ads
Ransomware	Locks files and demands ransom
Rootkits	Gains administrator-level (root) access by hiding malicious processes